

Welcome to

SC1006

Computer Organisation and Architecture

Assoc. Prof. Mohamed M. Sabry Aly

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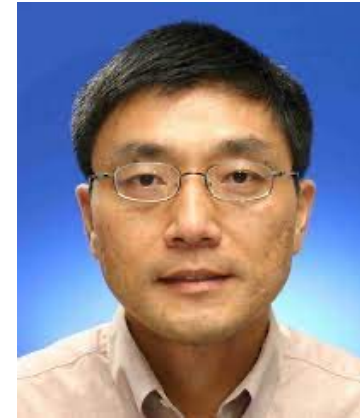


Instructors

- **Assoc/P Mohamed M. Sabry Aly** (Weeks #1 to #7)
- **Prof. Lin Weisi** (Weeks #8 to #13)



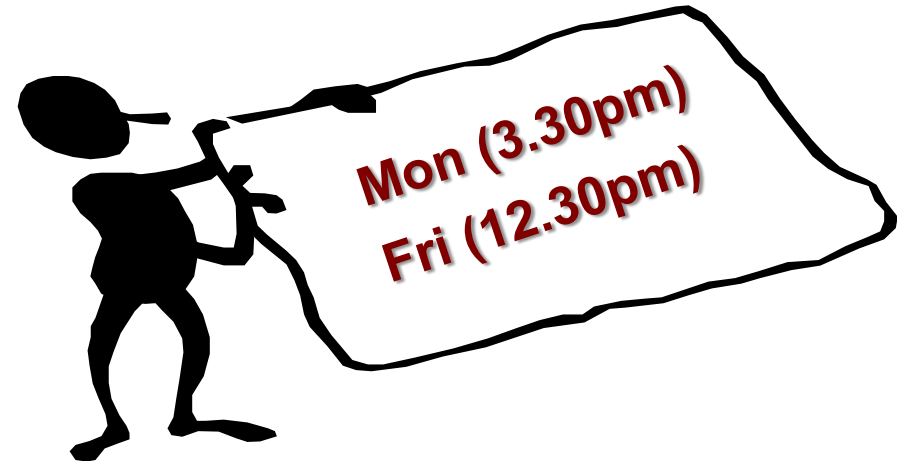
Mohamed Aly
(Course Coordinator)



Prof. Lin Weisi

Course Details

- **Review Sessions** - 13 weeks (LT4), 2 hours per week
- **Tutorials** - 12 weeks (1 hr per week, starting **week #2**)
- **Labs** - 4 experiments (2 hrs each), starting in **week #5 or #6**



Laboratory

1. 4 Experiments (about 1hr 45mins).
2. Lab #1 starts in week **#5** (for odd group)
3. Short **10 minute quiz** at the end of each lab.
4. Each student must keep record of their observations and solutions
 - Jot down results, observations and analysis into your lab manual.
 - Annotations may be useful references during quiz.
 - Properly done annotations can be submitted directly online as **lab assignments**.
5. Lab manuals available on the NTULearn ([Content/Laboratory](#))
6. Lab is in the Hardware Lab 2 at **N4-01b-05**

Laboratory Quiz

1. Multiple Choice Question (MCQ) Quiz.
 - Taken about 15 minutes before **end of lab**.
 - MCQ questions to be done in **10 mins**.
 - **Online quiz** done using NTULearn assessment tools.
2. Some useful information about the Lab Quiz:
 - A mix of simple and challenging questions on the **topics covered** by the lab experiment.
 - **Pace** yourself well during the quiz and you are given a **strict time limit** of 10 minutes.
 - Quiz is **open book**. You are free to use available tools used in the lab sessions... but watch your time!

Course Assessment

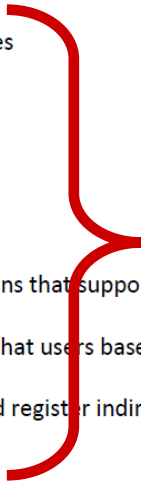
1. Lab Assignments (4 labs) – **40%**
2. Mid semester E test – **30%**
3. Final tests – **30%**

Review Sessions

- **Online review sessions**
 - Twice per week – Mondays (3.30pm) & Fridays (12.30pm).
 - Student **must** watch assigned **video lectures** before schedule review sessions.

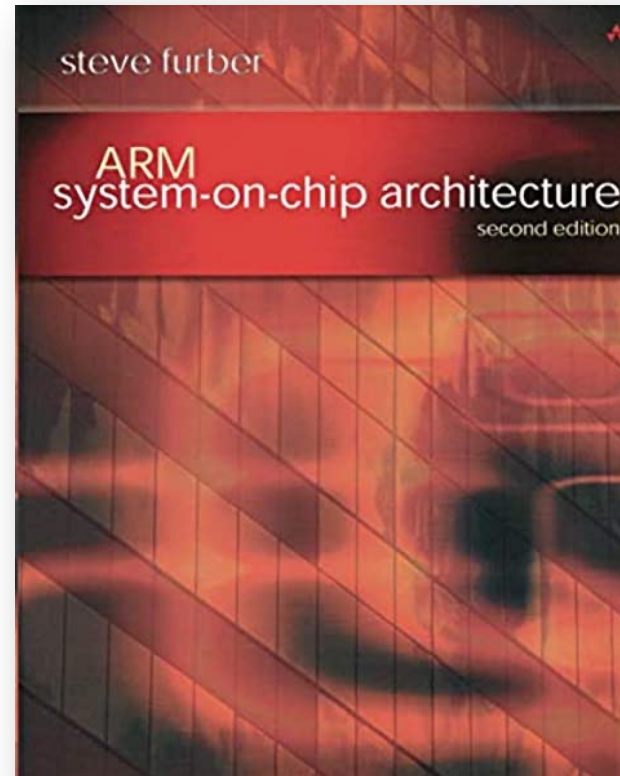
**Monday's
Review
Session in
Week #3**

4. Addressing Modes

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- (Week 3a)
- Video
Lectures
to watch**
1. Introduction to Assembly Programming and Addressing Modes
 - Identify why and when to use assembly programming.
 - Describe what are addressing modes.
 2. Register Direct and Immediate Addressing
 - Describe what is register direct.
 - Describe what is immediate data and its application.
 3. Register Indirect with Base Register
 - Describe what is register indirect and the ARM instructions that support this addressing mode.
 - Describe the variant and application of register indirect that uses base plus offset and index register.
 - Compare the relative pros and cons of register direct and register indirect addressing modes.
- (Week 3b)
4. Register Indirect with Autoindexing and Stacks
 - Describe what is the autoindexing feature of ARM's register indirect addressing mode.
 - Describe the differences between pre-index and post-index addressing modes.
 - Describe the various stack implementations and operations using the ARM addressing modes.
 5. PC-related Addressing Modes
 - Describe the difference between absolute and relative jump.
 - Describe the concept of position-independent code and how it is achieved.
 - Describe how data is accessed using PC-relative addressing.

Video Lecture Schedule

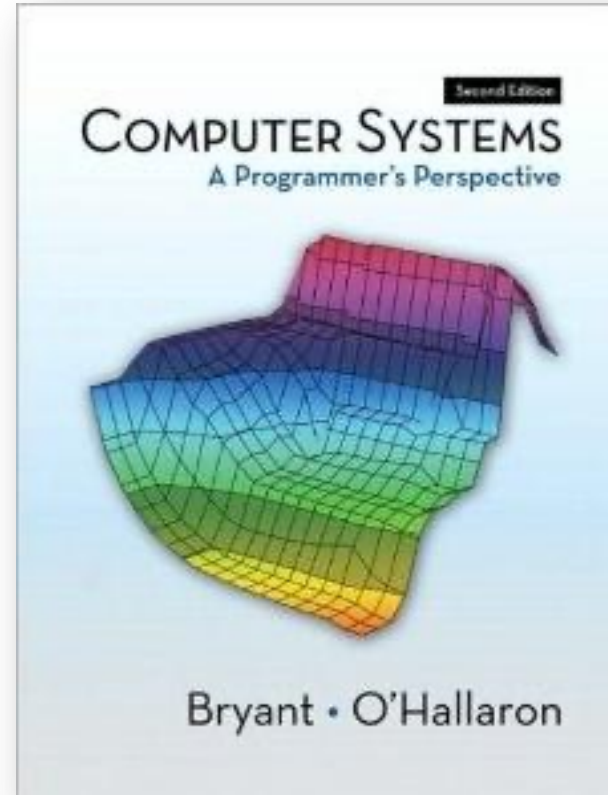
Reference Text



Steve Furber,
ARM system-on-chip architecture,
Addison-Wesley, 2000

Google: "Steve Furber ARM system-on-chip architecture"

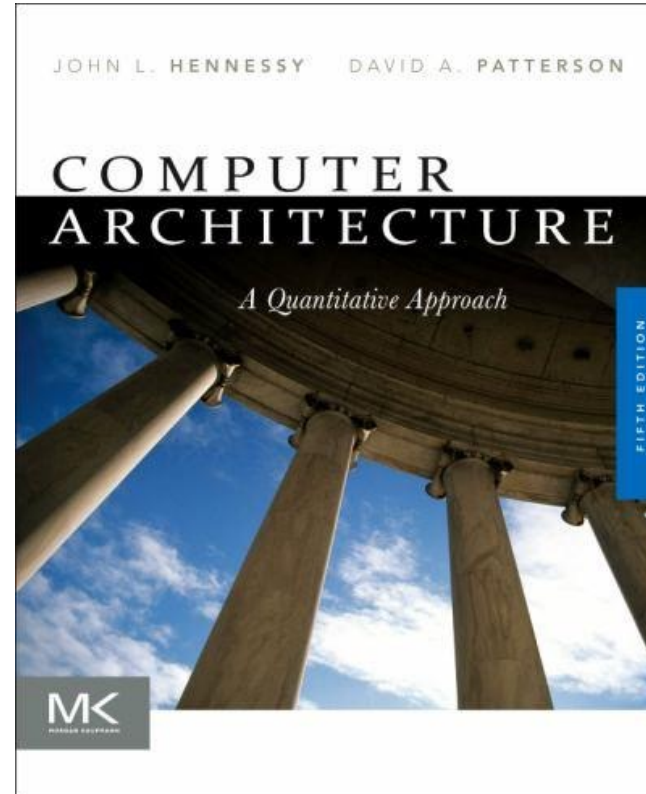
Reference Text



Randal E. Bryant & David O'Hallaron,
Computer Systems – A Programmer's Perspective,
Pearson Education International (2nd Edition), 2010.

CE1106/CZ1106

Reference Text



John L. Hennessy & David A. Patterson,
Computer Architecture – A Quantitative Approach,
Morgan Kaufmann (5th Edition), 2011.

Brief Course Outline and Schedule

1. Introduction, Data & Instruction Organization (wk 1-2)
2. Addressing Modes & Instruction Set (wk 3-4)
3. Modular and High-Level Programming (wk 5-7)
- Recess (1 week)
4. Signal-chain subsystem, DMA and Interrupts (wk 8)
5. Computer Memory (wk 9-10)
6. Comms and UI, Processor Pipeline (wk 11-12)
7. Computer Arithmetic and Review (wk 12-13)